

ORF 245 – Midterm Exam

Spring 2025

- Exam Duration: 80 minutes.
- Length: 4 Questions, all equally weighted. Be sure to carefully read each problem.
- **All of your answers (and supporting work) must be uploaded directly to Gradescope.** You may write your exam on a tablet, or in paper to scan and upload after the exam has ended. Please be sure to clearly box your final answers to each (sub-)question, and properly assign pages of your work to each question in Gradescope.

We will not be collecting any paper submissions for this exam. Before leaving, ensure that you confirm with the instructor or one of the TA's that we see your full exam submission on Gradescope.

- This exam is both open-note and open-book. You may reference class notes, the textbook, previous homework, precept notes, previous exams, etc. Anything posted on Canvas, or obtained during lecture is fair game. You may also use a calculator or R for performing computations.

You may access these resources electronically, subject to the honor code as stipulated below.

- You may **not** use any other online or external materials.
- Collaboration is (of course) not permitted. You should not discuss the exam questions with others until grades are released and solutions have been posted.
- Unless stated otherwise, please ensure you show and justify all important steps in deriving your answers.
- Please write out and sign the following pledge on the first page of your exam:

"I pledge my honor that I have not violated the Honor Code or the rules specified by the instructor during this examination."

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1. **(Conditional Probability; 25 pts)**

Suppose you have three email accounts, with the following properties:

Account	Messages Received (% of inbox)	Spam (% of each account)
1	70%	1%
2	20%	2%
3	10%	5%

We randomly select one email from our inbox, such that all emails are equally likely to be selected.

- (a) What is the probability that this email is from Account 1?
- (b) What is the probability that this email is spam?
- (c) What is the conditional probability that this email is from Account 1, *given that it is spam*?
- (d) What is the conditional probability that this email is from Account 1, *given that it is not spam*?
- (e) Define two events: $A = \{\text{selected email is from Account 1}\}$, and $B = \{\text{selected email is spam}\}$. Are the two events A and B independent? Justify your answer.

2. **(Discrete Random Variables; 25 pts)**

You and a friend are playing a simple betting game. Given a fair coin (which is equally likely to land heads- or tails-up), you flip it however many times needed until the first heads appears, and record the total number of flips as the random variable X .

- (a) What is the distribution of X ? Be sure to specify: i) the family of distribution, ii) the relevant parameters, and iii) the probability mass function of X , $p_X(\cdot)$.
- (b) Compute $\mathbb{P}(X > k)$ for $k = 1, 2, \dots$
- (c) Define a new random variable Y , which represents your payoff in this game.

$$Y = \begin{cases} 10\$ & \text{if } X \leq 3 \\ -1\$ & \text{otherwise.} \end{cases}$$

Compute $p_Y(\cdot)$, the probability mass function of Y .

- (d) Compute the expected payoff of this game, $\mathbb{E}[Y]$.

(e) Let us re-define the payoff random variable Y to be:

$$Y = \begin{cases} 10\$ & \text{if } X \leq 3 \\ -c\$ & \text{otherwise.} \end{cases}$$

for some constant c .

For what value of c is this game ‘fair’? That is, what value of c yields $\mathbb{E}[Y] = 0$?

3. (Continuous Random Variables; 25 pts)

The Pareto distribution, whose probability density function is parameterized by $\alpha > 1$, is given by:

$$f_X(x; \alpha) = \frac{\alpha}{x^{\alpha+1}}, \quad \text{for } x \geq 1. \quad (1)$$

Let $X \sim \text{Pareto}(\alpha)$. (i.e., the density of X is given by Eq. (1) for some value of α)

- (a) Verify that $f_X(x; \alpha)$ is a valid probability distribution function.
- (b) Compute the cumulative distribution function of X , $F_X(x; \alpha)$.
- (c) Compute $\mathbb{E}[X]$. Recall that $\alpha > 1$.

Hint: this should depend on α .

- (d) For $\alpha = 4$, compute $\text{Var}(X)$.
 - (e) Compute $\text{Var}(X)$ for general choice of $\alpha > 1$.
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4. (Jointly Distributed Random Variables; 25 pts)

Let X and Y be continuous random variables, whose joint probability density function is given by

$$f(x, y) = \begin{cases} c(x^2 + y^3) & \text{for } x \in [0, 1], y \in [0, 1] \\ 0 & \text{otherwise.} \end{cases}$$

for some constant c .

- (a) For what value of c is $f(x, y)$ a valid joint probability density function?
- (b) Compute the marginal probability density functions of X and Y , denoted $f_X(x)$ and $f_Y(y)$.
- (c) Are X and Y independent random variables? Justify your answer.
- (d) Compute $\text{Cov}(X, Y)$, the covariance between X and Y .
- (e) Compute $\rho(X, Y)$, the correlation between X and Y .

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